TAMILSELVAN.S

Full-Stack Web Developer & Python / Django Software Engineer

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PROFESSIONAL SUMMARY

Results-driven **Full-Stack Web Developer** and **Python / Django Software Engineer** with **8 months of professional experience** designing, engineering, and shipping high-performance digital products, automated cybersecurity countermeasures, and computer-vision-powered platforms. Proven ability to architect end-to-end applications: from crafting high-conversion, accessible user interfaces (**HTML5**, **CSS3**, **JavaScript ES6+**, **React.js**, **Bootstrap 5**, **Tailwind CSS**) to engineering resilient backend services (**Python 3.10+**, **Django 4.2**, **Django REST Framework / DRF**, **RESTful APIs**). Demonstrated specialization in real-time biometric face detection algorithms (**OpenCV**, **Haar Cascades**) and native Windows operating system endpoint security automation (**Win32 API**, **Windows ACLs**). Dedicated to clean architecture, normalized **SQL** database design (**SQLite**, **MySQL**), version control workflows (**Git/GitHub**), and cloud deployment lifecycle management (**Vercel**).

TECHNICAL SKILLS

PROGRAMMING LANGUAGES	**Python 3.10+**, **JavaScript (ES6+)**, **SQL**, **HTML5**, **CSS3**
WEB FRAMEWORKS & APIS	**Django 4.2**, **Django REST Framework (DRF)**, **React.js** (Basic Knowledge & Components), **RESTful APIs**, **MVT Architecture**, **Bootstrap 5**, **Tailwind CSS**, **WordPress**
DATABASES & STORAGE	**SQL**, **SQLite**, **MySQL**, **Django ORM**, **Relational Database Modeling**, **Schema Migrations**, **CRUD Operations**
COMPUTER VISION & AI	**OpenCV (cv2)**, **Haar Cascade Classifiers**, **Pearson Correlation Coefficient**, **Convolutional Neural Networks (CNN)**, **Deep Learning**, **HTML5 Webcam API**
OS & SECURITY ENGINEERING	**Windows Win32 API**, **Windows Security Event Logging (Event ID 4625)**, **Windows ACLs (icacls)**, **Role-Based Access Control (RBAC)**
TOOLS & CLOUD DEPLOYMENT	**Git**, **GitHub Version Control**, **Vercel Cloud Platform**, **Vite**, **Postman**, **Windows & Linux CLI**, **Mobile-First UI Architecture**

PROFESSIONAL EXPERIENCE

Full-Stack Web & Python Software Developer

Independent Software Engineering & Client Solutions | Dharmapuri, Tamil Nadu, India

`March 2026 - October 2026 (8 Months)`

Completed **8 months** of production-focused software development, engineering resilient full-stack web platforms, Python / Django REST APIs, and computer vision systems. All projects below were architected, developed, and deployed during this 8-month tenure:

- **Production Platform Architecture:** Architected, developed, and shipped **10+ full-stack web applications and digital platforms** across diverse commercial domains including automotive marketplaces, cosmetic healthcare clinics, travel booking portals, and corporate event management systems.
- **Computer Vision & Biometric Integration:** Engineered automated biometric verification pipelines combining **Python**, **Django REST Framework (DRF)**, and **OpenCV Haar Cascade Classifiers** with Pearson correlation similarity matching, achieving real-time employee attendance tracking with zero proxy clock-ins.
- **Automated Endpoint Defense & Security:** Built native Windows endpoint defense solutions actively monitoring Windows Security Event Logs (**Event ID 4625**), triggering automated Win32 workstation locking, and dynamically locking directory trees via **Windows ACLs (icacls)**.
- **Responsive Frontend Modernization:** Standardized high-fidelity responsive frontend architectures utilizing **HTML5**, **CSS3**, **Tailwind CSS**, **Bootstrap 5**, modern **JavaScript (ES6+)**, and foundational **React.js** components, ensuring 100% mobile responsiveness and sub-second rendering.
- **Database Modeling & Optimization:** Designed normalized relational database schemas in **SQLite** and **MySQL**, implementing robust migrations, indexing, and clean **Django ORM** queries.
- **Cloud Deployment & CI/CD:** Maintained continuous version control pipelines with **Git/GitHub** and managed cloud hosting and production deployments via the **Vercel Cloud Platform**.

KEY PROJECTS

Biometric Facial Recognition Attendance System

COMPUTER VISION FLAGSHIP

July – August 2026

Enterprise Attendance & Shift Analytics Platform with Real-Time Face Matching

- **Real-Time Facial Recognition Pipeline:** Integrated browser-based HTML5 webcam streaming with backend **OpenCV Haar Cascade Classifiers** for face detection and **Pearson Correlation Coefficient** similarity matching against stored biometric templates.
- **Anti-Spoofing & Shift Integrity:** Engineered automated biometric verification during shift clock-in and clock-out events, recording exact timestamps, face match confidence scores, and shift integrity flags to eliminate proxy attendance.
- **3-Tier RBAC Architecture:** Built standalone portals for Super Admin (global policies, match thresholds), Administrator (employee directories, shift rosters), and Employee (self-service clock-in, history).
- **Reporting & CSV Pipelines:** Automated daily late-arrival calculations, overtime tracking, and one-click monthly CSV report generation.

Technologies: `Python 3.10+` `Django 4.2` `Django REST Framework` `OpenCV (cv2)` `Haar Cascades` `HTML5 Webcam API` `SQLite` `Bootstrap 5` `CSV Reporting`

GitHub Repository ↗

Laptop Security – Real-Time Endpoint Defense System

CYBERSECURITY FLAGSHIP

September 2026

Native Python, Django & Computer Vision Endpoint Defense Platform

- **Automated Intrusion Detection:** Actively monitors Windows Security Event Logs (**Event ID 4625** for logon failures) utilizing a sliding-window counter to detect unauthorized access attempts.
- **Forensic Camera Capture & Auto-Lock:** Triggers native Win32 `LockWorkStation()` on breach, silently captures forensic photos via **OpenCV**, and transmits photographic evidence to an audit console.
- **Confidential Folder Vault:** Programmatically manages Windows ACL permissions (**icacls**) and system attribute flags (`attrib +h +s`) to lock and hide confidential directory trees, backed by Master Emergency PIN recovery.
- **Web Security Operations Center (SOC):** Built a live Django dashboard displaying real-time CPU/RAM/Disk telemetry, intruder photograph galleries, and emergency lockdown triggers.

Technologies: `Python 3.10` `Django` `OpenCV (cv2)` `Windows Win32 API` `Windows ACLs (icacls)` `Security Event Log API`

GitHub Repository ↗

Used Car Marketplace Platform

LATEST RELEASE (OCT 20, 2026)

October 20, 2026

Multi-Page Automotive Marketplace with Dual Portals & Financing Engine

- **Dual-Role Portal Architecture:** Developed standalone Admin Dashboard (dealership analytics, listing approvals, lead management) and User Dashboard (saved vehicles, test-drive appointments, offer tracking).
- **Inventory & Valuation Engine:** Built interactive vehicle search filtering by make, model, price, mileage, and body style paired with an automated *"Sell Your Car"* instant appraisal tool.
- **Inspection & Financing Systems:** Integrated comprehensive multi-point vehicle inspection report displays and dynamic EMI loan calculation tools with real-time interest adjustments.
- **Modern Automotive Design:** Crafted an ultra-sleek UI featuring **Tailwind CSS**, modern JavaScript, dark mode support, and interactive 3D perspective cards.

Technologies: `Tailwind CSS` `JavaScript ES6+` `Dual Admin/User Portals` `Automotive Valuation Engine` `EMI Calculator` `Mobile-First UI`

Vercel Live Demo ↗ GitHub ↗

Dermatology & Cosmetic Skin Clinic Platform RECENT RELEASE

October 16, 2026

Luxury Healthcare Web Application with Appointment & Treatment Engine

- **Dual-Role Clinic Dashboards:** Architected complete Admin Dashboard (patient rosters, revenue metrics, appointment queues) and Patient Portal (treatment history, doctor notes, upcoming sessions).
- **Interactive Before & After Image Sliders:** Implemented hardware-accelerated interactive image comparison sliders demonstrating clinical transformation results.
- **Procedures & Physician Directories:** Designed structured directories for medical procedures with physician bio cards and instant booking modal workflows.

Technologies: `Tailwind CSS` `JavaScript ES6+` `Before & After Slider UI` `Admin / Patient Portals` `Modal Workflows`

Vercel Live Demo ↗ GitHub ↗

Travel Agency & Holiday Packages Portal RECENT RELEASE

October 16, 2026

Modern Vacation Discovery & Tour Reservation Platform

- Engineered interactive tour catalogs for domestic destinations, international tours, and family vacation packages with real-time pricing and availability calculators.
- Implemented user reservation dashboard tracking travel confirmations, booking history, and downloadable travel guides.
- Crafted responsive, high-performance UI with smooth scroll transitions, mobile drawer navigation, and accessible forms.

Technologies: `HTML5 / CSS3` `JavaScript ES6` `User & Admin Dashboards` `CSS Flexbox / Grid` `SVG Vector Systems`

Vercel Live Demo ↗ GitHub ↗

Corporate Event Planning & Conference Management Firm RECENT RELEASE

October 15, 2026

Enterprise Event Scheduling & Delegate Ticketing UI

- Built multi-track conference schedule displays with speaker bios, session tracks, and venue locator integration.
- Streamlined attendee registration flows with form validation, package selection, and confirmation receipts.

Technologies: `HTML5 / Modern CSS` `Modular JavaScript` `Conference Schedules` `Form Validation`

Vercel Live Demo ↗ GitHub ↗

Additional Production Web Applications & Prototypes

September - October 2026

Commercial Client Solutions & React Educational Prototypes

- **Bagel Bakery & Specialty Café Platform (Oct 8, 2026):** Artisanal bakery storefront featuring interactive daily menu catalogs, online catering orders, image galleries, and responsive visit booking. (**HTML5**, **CSS3**, **JavaScript**).
- **Local Gym & Fitness Center Platform (Sep 2026):** Interactive gym web app featuring tiered membership selectors, class rosters, trainer directories, and interactive motion animations. (**Vanilla JS**, **motion.js**).
- **Kids Martial Arts Academy Portal (Oct 7, 2026):** Youth athletic academy website highlighting confidence-building programs, belt-progression curriculum, and trial class booking. (**Responsive UI**, **CSS Grid**).
- **Modern Course Learning Hub (React 19) (Sep 2026):** Single-page modern educational application built with **React 19** and **Vite** featuring modular course browsing components.

Technologies: `React 19` `Vite` `JavaScript ES6+` `motion.js` `CSS Flexbox & Grid`

View All Projects on Vercel ↗ View All Repositories on GitHub ↗

TECHNICAL PAPER PRESENTATION

Human Stress Detection Using Convolutional Neural Networks: An In Depth Analysis

Research Presentation | Deep Learning & Computer Vision Telemetry

`PAPER PRESENTATION`

Presented technical research conducting an in-depth analytical evaluation of **Convolutional Neural Network (CNN)** architectures applied to automated human stress detection and cognitive load classification:

- **CNN Deep Learning Architecture:** Investigated multi-tier convolutional filter feature mapping, max-pooling layers, and softmax classification trained on facial emotion landmarks, eye-blink dynamics, and micro-expression datasets.
- **Real-Time Facial Stress Telemetry:** Evaluated real-time camera feed inferencing pipelines to detect elevated stress indicators across varying illumination, head poses, and occlusion conditions.
- **Accuracy & Performance Benchmarking:** Conducted comparative analysis against conventional machine learning baselines (SVM, Random Forest), demonstrating the superior feature extraction accuracy of deep convolutional networks.

Core Competencies: `Convolutional Neural Networks (CNN)` `Deep Learning` `OpenCV` `Facial Landmark Telemetry` `Cognitive Load Analysis`

GitHub Research Code & Presentation ↗

CERTIFICATIONS & COURSE COMPLETION

Python Full Stack Development Certification

Comprehensive Full-Stack Web Architecture & Engineering

`Course Completed`

Successfully completed intensive practical training in **Python Full Stack Development**, mastering end-to-end web engineering from responsive frontend design and server-side Django frameworks to relational SQL databases and RESTful API integrations:

Backend & API: Python core programming, **Django Web Framework**, **Django REST Framework (DRF)**, MVT architecture, session security & authentication.

Frontend UI: HTML5, **CSS3**, modern **JavaScript (ES6+)**, **Bootstrap 5** responsive grid systems, and foundational **React.js**.

Databases & SQL: Relational database design, **SQL** querying, CRUD operations, **SQLite/MySQL**, and **Django ORM** integration.

Deployment & Version Control: Git/GitHub workflows, full-stack application lifecycle, and responsive multi-device testing.

EDUCATION

Master of Computer Applications (MCA)

Hindustan College Of Arts And Science

2023 - 2025 `CGPA: 7.62`

B.Sc. Computer Science

K.S. Rangasamy College of Arts and Science (Autonomous)

2020 - 2023 `CGPA: 7.271`

Higher Secondary Certificate (HSC)

Sri Vidya Mandir Matric Hr. Sec. School, Dharmapuri

2019 - 2020 `Percentage: 64%`

Secondary School Leaving Certificate (SSLC)

St. Paul's Matric Hr. Sec. School, Dharmapuri

2017 - 2018 `Percentage: 68%`